

Statutes of Graphs *

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Abstract

In this paper we give the sharp upper bound and lower bound for the status of a connected graph of order n and maximum degree k . The graphs attaining these bounds are also considered.

1 Introduction

All graphs considered in this paper are finite, simple, and without loops. Suppose that G is a connected graph. For a vertex x in G , the *status* of x is defined by

$$s_G(x) = \sum_{y \in V(G)} d_G(x, y),$$

where $d_G(x, y)$ denotes the distance between x and y . The *status* of the graph G is defined by

$$s(G) = \min_{x \in V(G)} s_G(x).$$

The *median* of G is the set of vertices in G with the minimum status $s(G)$.

Let T be a tree and x be a vertex in T . A *branch* at x is a maximal subtree of T which contains x as an endvertex. The *branch-weight* of x is the maximum number of edges in any branch at x . Equivalently, the branch-weight of x is the maximum number of vertices in any component of $T - x$. The *centroid* of T is the set of vertices in T with the minimum branch-weight. B. Zelinka [4] showed that any tree has its median equal to its

centroid. Statutes and medians are widely studied for trees (e.g. [1], [4], [5], [7], [8]) and general graphs (e.g. [2], [3], [6], [8], [9]).

Proposition 1.1 [1; 5, Lemma 3.1] *Let T be a tree and x be a vertex in T . Then x is in the centroid of T if and only if $|V(T')| \leq \frac{1}{2}|V(T)|$ for every component T' of $T - x$.* $\square$

Proposition 1.2 [1; 5, Theorem 3.1] *Any tree has its median equal to its centroid.* $\square$

Combining propositions 1.1 and 1.2, we have

Proposition 1.3 *Let T be a tree and x be a vertex in T . Then x is in the median of T if and only if $|V(T')| \leq \frac{1}{2}|V(T)|$ for every component T' of $T - x$.* $\square$

In this paper, we consider the range of $s(G)$ if G is a connected graph of order n and maximum degree $\Delta(G) = k$.

2 Bounds of $s(G)$ w.r.t $\Delta(G)$

Let T be a rooted tree with root z_0 . Then the *height* of T is defined by

$$\text{height}(T) = \max_{x \in V(T)} d_T(z_0, x).$$

Let T be a rooted tree with root z_0 and $\text{height}(T) \geq 2$. Suppose k is an integer ≥ 2 . If $\deg_T x = k$ whenever $x \in V(T)$, $d(z_0, x) \leq \text{height}(T) - 2$, then T is called a *balanced k -tree*. By Proposition 1.3, the root of a balanced k -tree T is contained in the median of T . If $\deg_T x = k$ whenever $x \in V(T)$, $d_T(z_0, x) \leq \text{height}(T) - 1$, then T is called a *full k -tree*. Obviously, a full k -tree is a balanced k -tree.

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Let $N_{k,h}$ denote the number of vertices of a full k -tree with height h . Then

$$\begin{aligned} N_{k,h} &= 1 + k + k(k-1) + k(k-1)^2 + \cdots \\ &\quad + k(k-1)^{h-1} \\ &= \begin{cases} 2h+1 & \text{if } k=2, \\ 1 + k \frac{(k-1)^h - 1}{k-2} & \text{if } k \geq 3. \end{cases} \end{aligned}$$

The number of vertices of a balanced k -tree with height h is between $N_{k,h-1} + 1$ and $N_{k,h}$ (inclusively).

A balanced k -tree of order n is denoted by $B_{n,k}$. Note that $B_{n,k}$ may not be unique if it is not full. The graphs in Fig. 1. are examples of $B_{14,3}$.

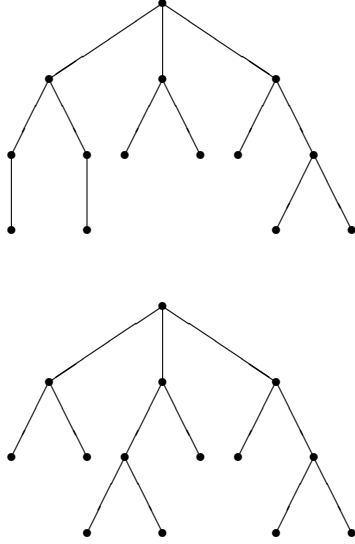


Fig. 1. examples of $B_{14,3}$

Suppose that $N_{k,h-1} + 1 \leq n \leq N_{k,h}$ for some integer $h \geq 2$. Then

$$\begin{aligned} s(B_{n,k}) &= k + 2k(k-1) + 3k(k-1)^2 + \cdots \\ &\quad + (h-1)k(k-1)^{h-2} + h \cdot (n - N_{k,h-1}). \end{aligned}$$

Let

$$\begin{aligned} b_{n,k} &= k + 2k(k-1) + 3k(k-1)^2 + \cdots \\ &\quad + (h-1)k(k-1)^{h-2} + h \cdot (n - N_{k,h-1}). \end{aligned}$$

For integers $n > k \geq 2$, the k -grass $G_{n,k}$ is the graph with $V(G_{n,k}) = \{x_1, x_2, \dots, x_n\}$ and $E(G_{n,k}) = \{x_i x_{i+1} : i = 1, 2, \dots, n-k\} \cup \{x_{n-k+1} x_j : j = n-k+2, n-k+3, \dots, n\}$.

We exhibit $G_{9,6}$ in Fig. 2. It is easy to see that $G_{n,2}$ is a path.

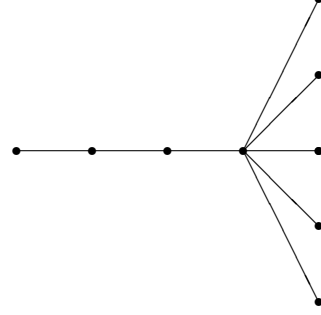


Fig. 2.

Note that $G_{n,k}$ has n vertices, and has a vertex with degree k , to be specific, $\deg_{G_{n,k}} x_{n-k+1} = k$.

We determine the status of $G_{n,k}$. Let $n \geq 3$ be an integer. Suppose that $2 \leq k \leq \lfloor \frac{n}{2} \rfloor$. By Proposition 1.3, we can see that $x_{\lfloor \frac{n}{2} \rfloor + 1}$ is in the median of $G_{n,k}$, and

$$\begin{aligned} s(G_{n,k}) &= s(x_{\lfloor \frac{n}{2} \rfloor + 1}) \\ &= 1 + 2 + \cdots + \lfloor \frac{n}{2} \rfloor + 1 + 2 + \cdots \\ &\quad + (n - k - \lfloor \frac{n}{2} \rfloor) \\ &\quad + (k-1)(n - k - \lfloor \frac{n}{2} \rfloor + 1) \\ &= \binom{\lfloor \frac{n}{2} \rfloor + 1}{2} + \binom{n - \lfloor \frac{n}{2} \rfloor - k + 1}{2} \\ &\quad + (k-1)(n - k - \lfloor \frac{n}{2} \rfloor + 1). \end{aligned}$$

Suppose that $\lfloor \frac{n}{2} \rfloor + 1 \leq k \leq n-1$. By Proposition 1.3, we can see that x_{n-k+1} is in the median of $G_{n,k}$, and

$$\begin{aligned} s(G_{n,k}) &= s(x_{n-k+1}) \\ &= 1 + 2 + \cdots + (n-k) + (k-1) \\ &= \binom{n-k+1}{2} + k-1. \end{aligned}$$

Thus $s(G_{n,k}) =$

$$\left\{ \begin{array}{l} \binom{\lfloor \frac{n}{2} \rfloor + 1}{2} + \binom{n - \lfloor \frac{n}{2} \rfloor - k + 1}{2} \\ + (k-1)(n-k - \lfloor \frac{n}{2} \rfloor + 1) \text{ if } 2 \leq k \leq \lfloor \frac{n}{2} \rfloor, \\ \binom{n-k+1}{2} + k-1 \text{ if } \lfloor \frac{n}{2} \rfloor + 1 \leq k \leq n-1. \end{array} \right.$$

Let $g_{n,k} = s(G_{n,k})$.

Lemma 2.1 *Let G be a connected graph. Suppose that x is a vertex of G . Then there exists a spanning tree T of G such that $s_T(x) = s_G(x)$.*

Proof. It is easy to see that there exists a spanning tree T of G such that $d_T(x, y) = d_G(x, y)$ for all y in $V(G)$. Then $s_T(x) = s_G(x)$. $\square$

Lemma 2.2 *Let $r \geq h$ be positive integers. Suppose that $\ell_1, \ell_2, \dots, \ell_r, p_1, p_2, \dots, p_h$ are positive numbers such that $\ell_1 \leq p_1, \ell_2 \leq p_2, \dots, \ell_{h-1} \leq p_{h-1}$ and $\ell_1 + \ell_2 + \dots + \ell_r = p_1 + p_2 + \dots + p_h$. Let $a_1 < a_2 < \dots < a_r$ be positive numbers. Then $a_1\ell_1 + a_2\ell_2 + \dots + a_r\ell_r \geq a_1p_1 + a_2p_2 + \dots + a_hp_h$, and the equality holds if and only if $r = h$ and $\ell_1 = p_1, \ell_2 = p_2, \dots, \ell_h = p_h$.*

Proof. We consider two cases: Case 1. $r = h$ and Case 2. $r \geq h + 1$.

Case 1. $r = h$. Then

$$\begin{aligned} & (a_1\ell_1 + a_2\ell_2 + \dots + a_r\ell_r) - \\ & (a_1p_1 + a_2p_2 + \dots + a_hp_h) \\ = & a_1(\ell_1 - p_1) + a_2(\ell_2 - p_2) + \dots \\ & + a_{h-1}(\ell_{h-1} - p_{h-1}) + a_h(\ell_h - p_h) \\ = & a_1(\ell_1 - p_1) + a_2(\ell_2 - p_2) + \dots \\ & + a_{h-1}(\ell_{h-1} - p_{h-1}) + a_h(p_1 + p_2 + \dots \\ & + p_{h-1} - \ell_1 - \ell_2 - \dots - \ell_{h-1}) \\ = & (a_1 - a_h)(\ell_1 - p_1) + (a_2 - a_h)(\ell_2 - p_2) + \dots \\ & + (a_{h-1} - a_h)(\ell_{h-1} - p_{h-1}) \\ \geq & 0. \end{aligned}$$

Thus $a_1\ell_1 + a_2\ell_2 + \dots + a_r\ell_r \geq a_1p_1 + a_2p_2 + \dots + a_hp_h$, and the equality holds if and only if $\ell_1 = p_1, \ell_2 = p_2, \dots, \ell_h = p_h$.

Case 2. $r \geq h + 1$. Then

$$\begin{aligned} & (a_1\ell_1 + a_2\ell_2 + \dots + a_r\ell_r) \\ & - (a_1p_1 + a_2p_2 + \dots + a_hp_h) \\ > & (a_1\ell_1 + a_2\ell_2 + \dots + a_h\ell_h + a_h\ell_{h+1} + \dots \\ & + a_h\ell_r) - (a_1p_1 + a_2p_2 + \dots + a_hp_h) \\ = & a_1(\ell_1 - p_1) + a_2(\ell_2 - p_2) + \dots \\ & + a_{h-1}(\ell_{h-1} - p_{h-1}) \\ & + a_h(\ell_h + \ell_{h+1} + \dots + \ell_r - p_h) \end{aligned}$$

$$\begin{aligned} = & a_1(\ell_1 - p_1) + a_2(\ell_2 - p_2) + \dots \\ & + a_{h-1}(\ell_{h-1} - p_{h-1}) + a_h(p_1 + p_2 + \dots + p_{h-1} \\ & - \ell_1 - \ell_2 - \dots - \ell_{h-1}) \\ = & (a_1 - a_h)(\ell_1 - p_1) + (a_2 - a_h)(\ell_2 - p_2) + \dots \\ & + (a_{h-1} - a_h)(\ell_{h-1} - p_{h-1}) \\ \geq & 0. \end{aligned}$$

Thus $a_1\ell_1 + a_2\ell_2 + \dots + a_r\ell_r > a_1p_1 + a_2p_2 + \dots + a_hp_h$. $\square$

The following lemma is trivial and we omit the proof.

Lemma 2.3 *Let M, N, p be positive integers with $M \leq N \leq p$. Suppose that $a_1, a_2, \dots, a_p$ are positive integers such that $\{a_1, a_2, \dots, a_p\} = \{1, 2, \dots, M\}$. Then $a_1 + a_2 + \dots + a_p \leq 1 + 2 + \dots + (N-1) + (p-N+1)N$.* $\square$

For a graph G and a vertex x in G , the *eccentricity* of x is defined by $e_G(x) = \max_{y \in V(G)} d_G(y, x)$. Now we prove the main result of this paper. Recall that $b_{n,k}$ and $g_{n,k}$ denote $s(B_{n,k})$ and $s(G_{n,k})$ respectively.

Theorem 2.4 *Suppose that G is a connected graph of order n and $\Delta(G) = k \geq 2$. Then*

- (1) $b_{n,k} \leq s(G) \leq g_{n,k}$,
- (2) $s(G) = b_{n,k}$ if and only if G contains some spanning $B_{n,k}$, and
- (3) if $s(G) = g_{n,k}$, then G contains a spanning $G_{n,k}$.

Proof. The theorem is trivial for $k = 2$. Assume that $k \geq 3$. We prove the theorem by proving the following: (1') $b_{n,k} \leq s(G)$, and the equality holds if and only if G contains a spanning $B_{n,k}$. (2') $s(G) \leq g_{n,k}$, and if the equality holds then G contains a spanning $G_{n,k}$.

Now we prove (1').

(1') Let m be a vertex in the median of G . By Lemma 2.1, G has a spanning tree T such that $s_T(m) = s_G(m) = s(G)$. Let $r = e_T(m)$. For $i = 1, 2, \dots, r$, let $L_i = \{x \in V(T) : d(x, m) = i\}$. Since $\Delta(G) = k$, we have $\deg_T x \leq k$ for all $x \in V(T)$. Hence

$$\begin{aligned} |L_1| & \leq k, \\ |L_2| & \leq |L_1|(k-1) \leq k(k-1), \\ |L_3| & \leq |L_2|(k-1) \leq k(k-1)^2, \\ & \vdots \end{aligned}$$

Thus for $i = 1, 2, \dots, r$, $|L_i| \leq k(k-1)^{i-1}$. Suppose that $N_{k,h-1} + 1 \leq n \leq N_{k,h}$ for some integer h . Obviously, $r \geq h$. Hence

$$\begin{aligned} s(G) &= s_T(m) \\ &= 1|L_1| + 2|L_2| + \dots + r|L_r| \\ &\geq k + 2k(k-1) + 3k(k-1)^2 \dots \\ &\quad + (h-1)k(k-1)^{h-2} + h \cdot (n - N_{k,h-1}) \end{aligned}$$

the last inequality follows from the fact that $|L_1| + |L_2| + \dots + |L_r| = n - 1 = k + k(k-1) + \dots + k(k-1)^{h-2} + (n - N_{k,h-1})$ and Lemma 2.2. Thus $s(G) \geq b_{n,k}$.

Now suppose that $s(G) = b_{n,k}$. From the proof of $s(G) \geq b_{n,k}$ and applying condition for the equality in Lemma 2.2, we see that $r = h$ and $|L_1| = k, |L_2| = k(k-1), |L_3| = k(k-1)^2, \dots, |L_{r-1}| = k(k-1)^{r-2}, |L_r| = n - N_{k,r-1}$. Hence T is a balanced k -tree $B_{n,k}$. Since T is a spanning tree of G , G contains a spanning $B_{n,k}$.

Conversely, suppose that G contains a spanning $B_{n,k}$. Then $s(G) \leq s(B_{n,k}) = b_{n,k}$. On the other hand, we just proved that $s(G) \geq b_{n,k}$. Thus $s(G) = b_{n,k}$.

Now we prove (2').

(2') Suppose that u is a vertex with $\deg_G u = k$. Let T be a spanning tree of G such that $uv \in E(T)$ for all $v \in N_G(u)$.

Let m be a vertex in the median of T . Consider two cases: Case 1: $u = m$, Case 2: $u \neq m$.

Case 1: $u = m$.

Let $\ell = e_T(u)$. Let y_ℓ be a vertex such that $d_T(u, y_\ell) = \ell$, and let $uy_1y_2 \dots y_\ell$ be the path joins u and y_ℓ in T . Consider two subcases, Case 1.a: $\lfloor \frac{n}{2} \rfloor + 1 \leq k \leq n - 1$, Case 1.b: $2 \leq k \leq \lfloor \frac{n}{2} \rfloor$.

Case 1.a: $\lfloor \frac{n}{2} \rfloor + 1 \leq k \leq n - 1$.

Since $N_T(u) - \{y_1\} \subset V(G) - \{u, y_1, y_2, \dots, y_\ell\}$, we have $k - 1 \leq n - \ell - 1$. Let all the $d_T(u, x)$, where $x \in V(T) - \{u, y_1, y_2, \dots, y_\ell\}$, be $d_1 \leq d_2 \leq \dots \leq d_{k-1} \leq \dots \leq d_{n-\ell-1}$. Since $\deg_T u = k$, we have $d_1 = d_2 = \dots = d_{k-1} = 1$. Hence

$$\begin{aligned} s(G) &\leq s(T) \\ &= s_T(m) \\ &= s_T(u) \\ &= d_T(u, y_1) + d_T(u, y_2) + \dots + d_T(u, y_\ell) \\ &\quad + \sum_{x \notin \{u, y_1, y_2, \dots, y_\ell\}} d_T(u, x) \\ &= 1 + 2 + \dots + \ell + d_1 + d_2 + \dots + d_{k-1} \\ &\quad + d_k + d_{k+1} + \dots + d_{n-\ell-1} \\ &= 1 + 2 + \dots + \ell + (k-1) \cdot 1 \end{aligned}$$

$$\begin{aligned} &\quad + d_k + d_{k+1} + \dots + d_{n-\ell-1} \\ &\leq 1 + 2 + \dots + \ell + (k-1) \\ &\quad + (\ell+1) + (\ell+2) + \dots + (n-k) \\ &= 1 + 2 + \dots + \ell + (\ell+1) + (\ell+2) + \dots \\ &\quad + (n-k) + (k-1) \\ &= g_{n,k}. \end{aligned}$$

Thus $s(G) \leq g_{n,k}$.

Suppose that $s(G) = g_{n,k}$. From the proof of $s(T) \leq g_{n,k}$, we see that the equality $s(G) = g_{n,k}$ implies that $\ell = n - k$ and $d_k, d_{k+1}, \dots, d_{n-\ell-1}$ do not exist, which implies that $T = G_{n,k}$. Hence G contains a spanning $G_{n,k}$.

Case 1.b. $2 \leq k \leq \lfloor \frac{n}{2} \rfloor$.

Now let T_1 be the component of $T - u$ such that $y_\ell \in V(T_1)$. Since u is in the median of T , by Proposition 1.3, $|V(T_1)| \leq \frac{1}{2}|V(T)|$, we have $\ell \leq |V(T_1)| \leq \lfloor \frac{n}{2} \rfloor$. Let all the $d_T(u, x)$, where $x \in V(T) - (\{u\} \cup V(T_1))$, be $d_1 \leq d_2 \leq \dots \leq d_s$, where $s = n - 1 - |V(T_1)|$. Since $|V(T_1)| \leq \lfloor \frac{n}{2} \rfloor$, we have $s \geq n - \lfloor \frac{n}{2} \rfloor - 1$. And since $\deg_T u = k$, we have $d_1 = d_2 = \dots = d_{k-1} = 1$. Now

$$\begin{aligned} s(G) &\leq s(T) \\ &= s_T(m) \\ &= s_T(u) \\ &= d_T(u, y_1) + d_T(u, y_2) + \dots + d_T(u, y_\ell) \\ &\quad + \sum_{x \in V(T_1) - \{y_1, y_2, \dots, y_\ell\}} d_T(u, x) \\ &\quad + \sum_{x \in V(T) - (V(T_1) \cup \{u\})} d_T(u, x) \\ &\leq (1 + 2 + \dots + \ell) + (|V(T_1)| - \ell)\ell \\ &\quad + (d_1 + d_2 + \dots + d_s) \\ &= (1 + 2 + \dots + \ell) + (|V(T_1)| - \ell)\ell \\ &\quad + d_1 + d_2 + \dots + d_{n-\lfloor \frac{n}{2} \rfloor - 1} \\ &\quad + d_{n-\lfloor \frac{n}{2} \rfloor} + \dots + d_s \\ &\leq (1 + 2 + \dots + \ell) + (|V(T_1)| - \ell)\ell \\ &\quad + d_1 + d_2 + \dots + d_{n-\lfloor \frac{n}{2} \rfloor - 1} \\ &\quad + (s - n + \lfloor \frac{n}{2} \rfloor + 1)\ell \\ &= 1 + 2 + \dots + \ell + (\lfloor \frac{n}{2} \rfloor - \ell)\ell + d_1 + d_2 \\ &\quad + \dots + d_{n-\lfloor \frac{n}{2} \rfloor - 1} \\ &\leq 1 + 2 + \dots + \ell + (\ell+1) + \dots + \lfloor \frac{n}{2} \rfloor \\ &\quad + d_1 + d_2 + \dots + d_{n-\lfloor \frac{n}{2} \rfloor - 1}. \end{aligned}$$

Here we give an upper bound for $d_1 + d_2 + \dots + d_{n-\lfloor \frac{n}{2} \rfloor - 1}$. Note that in this subcase, $2 \leq k \leq$

$\lfloor \frac{n}{2} \rfloor$; thus $k-1 \leq \lfloor \frac{n}{2} \rfloor - 1 \leq n - \lfloor \frac{n}{2} \rfloor - 1$. Let $M = d_{n-\lfloor \frac{n}{2} \rfloor-1}$. It is easy to see that $\{d_1, d_2, \dots, d_{k-1}, \dots, d_{n-\lfloor \frac{n}{2} \rfloor-1}\} = \{1, 2, \dots, M\}$. Since $d_1 = d_2 = \dots = d_{k-1} = 1$, we have $M \leq n - \lfloor \frac{n}{2} \rfloor - k + 1$. Let $N = n - \lfloor \frac{n}{2} \rfloor - k + 1$ and $p = n - \lfloor \frac{n}{2} \rfloor - 1$. Then $M \leq N \leq p$. Applying Lemma 2.3, we obtain

$$\begin{aligned} & d_1 + d_2 + \dots + d_{k-1} + \dots + d_{n-\lfloor \frac{n}{2} \rfloor-1} \\ & < 1 + 2 + \dots + (N-1) + (p-N+1)N \\ & = 1 + 2 + \dots + (n - \lfloor \frac{n}{2} \rfloor - k) \\ & \quad + (k-1)(n - \lfloor \frac{n}{2} \rfloor - k + 1). \end{aligned}$$

Thus

$$\begin{aligned} s(G) & \leq 1 + 2 + \dots + \lfloor \frac{n}{2} \rfloor \\ & \quad + d_1 + d_2 + \dots + d_{n-\lfloor \frac{n}{2} \rfloor-1} \\ & < 1 + 2 + \dots + \lfloor \frac{n}{2} \rfloor \\ & \quad + 1 + 2 + \dots + (n - \lfloor \frac{n}{2} \rfloor - k) \\ & \quad + (k-1)(n - \lfloor \frac{n}{2} \rfloor - k + 1) \\ & = g_{n,k}. \end{aligned}$$

Hence $s(G) < g_{n,k}$.

Case 2: $u \neq m$.

Let the components of $T-m$ be $T_1, T_2, \dots, T_j$, some $j \geq 2$ such that $u \in V(T_1)$.

Let $\ell = \max_{x \notin V(T_1)} d_T(m, x)$. Suppose that y_ℓ is a vertex such that $d_T(m, y_\ell) = \ell$. We may assume that $y_\ell \in V(T_2)$. Let $my_1y_2 \dots y_\ell$ be the path joining m and y_ℓ in T .

Let all the $d_T(m, x)$, where $x \in V(T) - (\{m\} \cup V(T_1) \cup V(T_2))$, be $d_1 \leq d_2 \leq \dots \leq d_s$ where $s = n - 1 - |V(T_1)| - |V(T_2)|$. Since m is in the median of T , again by Proposition 1.3, $|V(T_2)| \leq \lfloor \frac{n}{2} \rfloor$; thus $s \geq n - |V(T_1)| - \lfloor \frac{n}{2} \rfloor - 1$. Now

$$\begin{aligned} & s(G) \\ & \leq s(T) \\ & = s_T(m) \\ & = d_T(m, y_1) + d_T(m, y_2) + \dots + d_T(m, y_\ell) \\ & \quad + \sum_{x \in V(T_2) - \{y_1, y_2, \dots, y_\ell\}} d_T(m, x) \\ & \quad + d_1 + d_2 + \dots + d_s + \sum_{x \in V(T_1)} d_T(m, x) \\ & \leq (1 + 2 + \dots + \ell) + (|V(T_2)| - \ell)\ell \\ & \quad + d_1 + d_2 + \dots + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1} \\ & \quad + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor} + \dots + d_s \\ & \quad + \sum_{x \in V(T_1)} d_T(m, x) \end{aligned}$$

$$\begin{aligned} & \leq (1 + 2 + \dots + \ell) + (|V(T_2)| - \ell)\ell \\ & \quad + d_1 + d_2 + \dots + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1} \\ & \quad + (\lfloor \frac{n}{2} \rfloor - |V(T_2)|)\ell + \sum_{x \in V(T_1)} d_T(m, x) \\ & \leq 1 + 2 + \dots + \ell + (\ell + 1) + (\ell + 2) + \dots \\ & \quad + |V(T_2)| + (|V(T_2)| + 1) + \dots + \lfloor \frac{n}{2} \rfloor \\ & \quad + d_1 + d_2 + \dots + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1} \\ & \quad + \sum_{x \in V(T_1)} d_T(m, x). \end{aligned}$$

Now we give an upper bound for

$$d_1 + d_2 + \dots + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1} + \sum_{x \in V(T_1)} d_T(m, x).$$

Let $N_T(u) = \{z_1, z_2, \dots, z_k\}$ such that $d_T(z_1, m) = d_T(z_2, m) = \dots = d_T(z_{k-1}, m)$. Let $M = \max(\{d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1}\} \cup \{d_T(m, x) : x \in V(T_1)\})$. We see that $\{d_1, d_2, \dots, d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1}\} \cup \{d_T(m, x) : x \in V(T_1)\} = \{1, 2, \dots, M\}$.

Furthermore,

$$\begin{aligned} M & \leq n - |V(T_1)| - \lfloor \frac{n}{2} \rfloor - 1 + |V(T_1)| - (k-2) \\ & = n - \lfloor \frac{n}{2} \rfloor - k + 1. \end{aligned}$$

Let $N = n - \lfloor \frac{n}{2} \rfloor - k + 1$ and $p = n - \lfloor \frac{n}{2} \rfloor - 1$. Since there are p number of positive integers (multiplicities are counted) in set $\{d_1, d_2, \dots, d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1}\} \cup \{d_T(m, x) : x \in V(T_1)\}$, and $M \leq N \leq p$.

Applying Lemma 2.3, we have

$$\begin{aligned} & d_1 + d_2 + \dots + d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1} \\ & \quad + \sum_{x \in V(T_1)} d_T(m, x) \\ & \leq 1 + 2 + \dots + (N-1) + (p-N+1)N \\ & = 1 + 2 + \dots + (n - \lfloor \frac{n}{2} \rfloor - k) \\ & \quad + (k-1)(n - \lfloor \frac{n}{2} \rfloor - k + 1). \end{aligned}$$

Thus

$$\begin{aligned} s(G) & \leq 1 + 2 + \dots + \lfloor \frac{n}{2} \rfloor \\ & \quad + 1 + 2 + \dots + (n - \lfloor \frac{n}{2} \rfloor - k) \\ & \quad + (k-1)(n - \lfloor \frac{n}{2} \rfloor - k + 1) \\ & = g_{n,k}. \end{aligned}$$

The last equality follows from the fact that $k \leq |V(T_1)| \leq \lfloor \frac{n}{2} \rfloor$. Thus $s(G) \leq g_{n,k}$. Suppose that

$s(G) = g_{n,k}$. From the above proof of $s(G) \leq g_{n,k}$, we see that the equality $s(G) = g_{n,k}$ implies that

- (1) $V(T_2) - \{y_1, y_2, \dots, y_\ell\} = \emptyset$,
- (2) $d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor}, d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor+1}, \dots, d_s$ do not exist,
- (3) the values of $d_1, d_2, \dots, d_{n-|V(T_1)|-\lfloor \frac{n}{2} \rfloor-1}$ and $d_T(m, x) (x \in V(T_1))$ are $1, 2, \dots, n - \lfloor \frac{n}{2} \rfloor - k$ (each with multiplicity 1) and $n - \lfloor \frac{n}{2} \rfloor - k + 1$ (with multiplicity $k - 1$)

We see that (1) and (2) imply that T_2 is a path and $\ell = |V(T_2)| = \lfloor \frac{n}{2} \rfloor$. Also we see that (2) and (3) imply that $T_3, T_4, \dots, T_j$ do not exist, and T_1 is the k -grass which consists of the edges $uz_1, uz_2, \dots, uz_{k-1}$ and the path joining u and m . Thus $T = G_{n,k}$. Hence G contains a spanning $G_{n,k}$. $\square$

Consider the graphs F and G in Fig. 3. Both F and G are of order 7 and maximum degree 4. We have $s(F) = 8, s(G) = 9$ and $s(G_{7,4}) = 9$. The graph F shows that the converse of Theorem 2.4 (3) does not hold. The graph G shows that the conclusion of Theorem 2.4 can not be replaced by $G \cong G_{7,4}$.

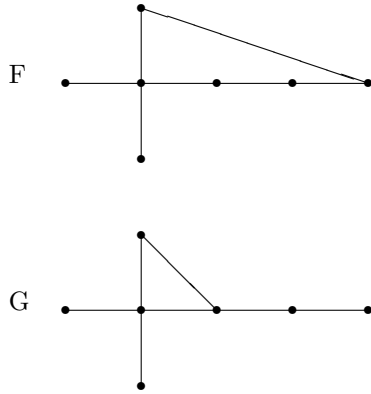


Fig. 3.

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